

REXplore: A Sketch based Interactive Explorer for Real Estates using Building Floor Plan Images

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Abstract—The increasing trend of using online platforms for real estate rent/sale makes automatic retrieval of similar floor plans a key requirement to help architects and buyers alike. Although sketch based image retrieval has been explored in the multimedia community, the problem of hand-drawn floor plan retrieval has been less researched in the past. In this paper, we propose REXplore (Real Estate eXplore), a novel framework that uses sketch based query mode to retrieve corresponding similar floor plan images from a repository using Cyclic Generative Adversarial Networks (Cyclic GAN) for mapping between sketch and image domain. The key contributions of our proposed approach are : (1) a novel sketch based floor plan retrieval framework using an intuitive and convenient sketch query mode; (2) A conjunction of Cyclic GANs and Convolution Neural Networks (CNNs) for the task of hand-drawn floor plan image retrieval. Extensive experimentation and comparison with baseline results authenticates our claim.

Keywords—Sketch; GAN; Floor Plan; Retrieval; Interactive

I. INTRODUCTION

The real estate industry is steadily moving towards using online platforms for various phases from planning, constructions, and sales. Interactive search techniques like sketches to retrieve similar floor plans can also benefit the architects and customers while searching for floor plans in the early design phases of a building. However, sketch based processing and retrieval has various challenges. Apart from being abstract and subjective in nature, there is a distinct difference in the sketch and image domain images. Recently, floor plan analysis and retrieval have been addressed by researchers [1], [2] in the image domain. In [3], a sketch-based similar floor plan retrieval approach was proposed to query the floor plans based on rooms, zones, units and levels. However, there was no end-to-end framework proposed in [3] and was not realistic. This motivated us to propose an interactive platform (system), where one can search for similar floor plan images based on hand-drawn sketches.

The existing scholarship on sketch recognition uses the conventional image classification paradigm with the help of hand-crafted features and supervised classifiers [4]. However, hand-crafted features traditionally used (such as HOG [5], SIFT [6], Run-Length Histogram [7] and Online Algorithm for Scalable Image Similarity (OASIS) [8]) are not able to capture well the abstract and sparse nature of

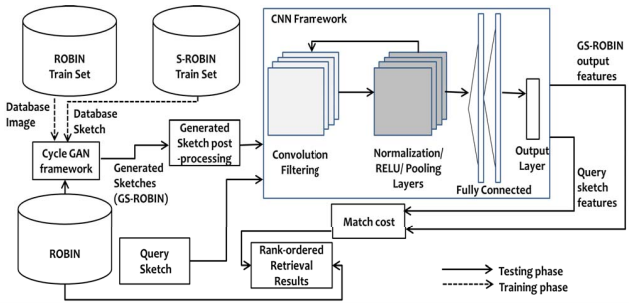


Figure 1. Schematic showing the architecture of REXplore.

sketches. Also, deep learning methods using Convolutional Neural Networks (CNNs) as proposed in [1] and [9] extract deep features from floor plans. However, they are not being able to capture the domain changes from sketch to image. Alternatively, Generative Adversarial Networks have also been employed for generic sketch retrieval [10], [11] but have not yet explored application in floor plan retrieval.

Hence, looking at the state-of-the art in floor plan retrieval, in this paper we propose REXplore (Real Estate eXplore), a novel floor plan retrieval framework. REXplore uses sketch as the query mode. REXplore is developed on Cyclic Generative Adversarial Networks (Cyclic GANs) for domain mapping. The key contributions of our approach are : (1) a novel sketch based interactive floor plan retrieval framework using an intuitive and convenient sketch query mode; (2) A conjunction of Cyclic GANs and CNNs for the task of domain mapping and hand-drawn floor plan retrieval.

Organization of the paper: In Sec. II, we describe the proposed deep learning architecture using Cyclic GAN. Section III describes the experimental setup and implementation details of this work. In Sec. IV, we give qualitative and quantitative analysis of the retrieval results. Finally, Sec. V concludes the paper.

II. METHODOLOGY

Here, we present the various steps in REXplore.

A. Domain Mapping through Cyclic GAN

The proposed Cyclic GANs based architecture learns to translate between the two domains, sketch and image, with-

out the explicit specification of paired input-output examples [12]. Here, our goal is to learn mapping functions between two domains, namely, S (sketch domain) and I (image domain). We have replicated the samples in ROBIN by sketch (henceforth will be referred as S-ROBIN). Given the training samples $\{s_a\}_{a=1}^n$ and $\{i_b\}_{b=1}^m$, in each domain, where, $s \in S$ and $i \in I$, n and m are the total number of samples in S-ROBIN and ROBIN, respectively. REXplore includes two mappings $G : S \rightarrow I$ and $F : I \rightarrow S$. Additionally, there are two adversarial discriminators D_S and D_I , where D_S aims to distinguish between sketches S and translated images $F(I)$; in the same way, D_I aims to discriminate between I and $G(S)$. The adversarial loss helps in matching the distribution of generated sketches to that in the target domain. The cycle consistency loss helps to prevent the learned mappings G and F from contradicting each other.

During training, we give unpaired input-output examples from both S-ROBIN and ROBIN that further learn to translate between domains. The assumption being, there is an underlying relationship between both the domains. The two losses applied to train the system are, adversarial loss ($\mathcal{L}_{GAN}$) and cycle consistency loss ($\mathcal{L}_{Cyc}$). Adversarial Loss has functions, which try to learn images similar to sketches and vice-versa. To ensure the mapping between sketch and image domain is cycle-consistent, a cycle consistency loss is used. This is followed by building a discriminator (a CNN), which takes an image as an input and tries to predict if it is an original or the output from the generator. The final objective function of the GAN network is:

$$\mathcal{L}(G, F, D_I, D_S) = \mathcal{L}_{GAN}(F, D_S, I, S) + \mathcal{L}_{GAN}(G, D_I, S, I) + \lambda \mathcal{L}_{Cyc}(G, F) \quad (1)$$

which is minimized over G and F , as well as maximized over D_I and D_S to train our Cyclic GAN framework.

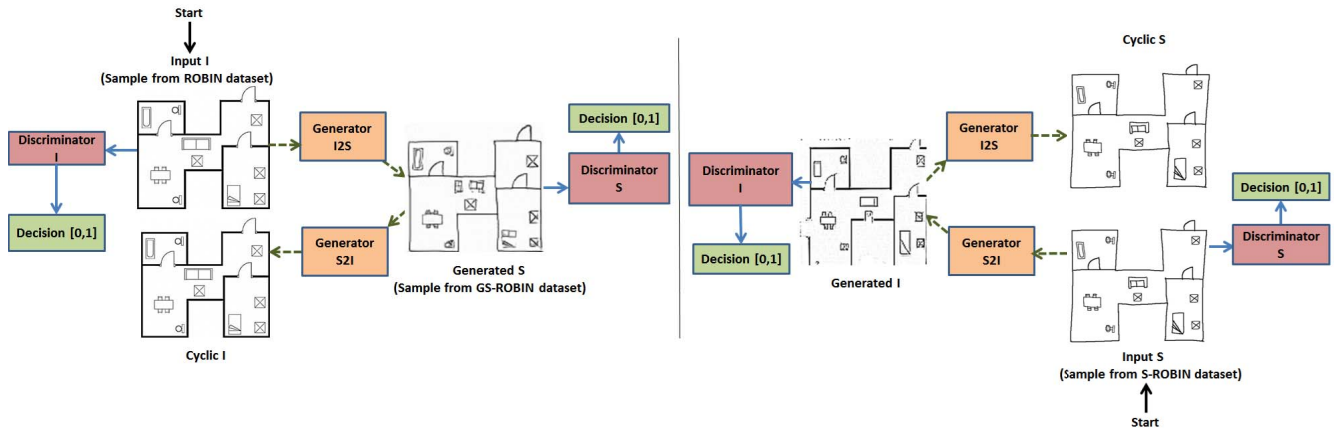


Figure 2. Network architecture for the Cyclic GAN framework, representing the inputs from ROBIN and S-ROBIN dataset. The outputs being generated constitute the GS-ROBIN dataset.

After this training and testing phase, we obtained a set of generated images corresponding to the sketches, as well as, a set of sketches corresponding to the ROBIN dataset (refer Fig. 2). We observed that if we take the obvious path, of converting our query sketches into corresponding image counterparts and then retrieve from the ROBIN dataset, the retrieval performance was quite weak giving an average precision value as low as 0.1631 and 0.1219, upon being trained on ROBIN as well as S-ROBIN dataset respectively. The reason being the images generated from sketches using the Cyclic GAN framework are of a poor quality than the sketches generated from the same framework. Therefore, we chose the path of converting our ROBIN dataset into sketches and then performing retrieval to yield better results. Thus, after this step we get a set of sketches (GS-ROBIN dataset) generated from the ROBIN-dataset which we use for retrieval purposes. To improve the quality of the generated sketches, we sharpen them as a post-processing step.

B. Feature Representation and Retrieval

In this work, we use a CNN with a combination of convolutional, pooling, normalization, drop out and fully connected layers, similar to described in the work [1], to obtain an effective feature representation from floor plans. For learning a representation of CNN (deep) layers, we experimented with training the deep model on the dataset. We further extracted the features from both the sketched query samples as well as the ROBIN generated sketches (GS-ROBIN dataset) used as the retrieval database.

After feature representation, retrieval of the images is done as follows: Let $(x_\phi^{GS}, y_\phi^{GS})$, $\phi = 1, 2, \dots, m$, represent the generated sketches, where, m is the total number of samples in the GS-ROBIN dataset, x_ϕ^{GS} is the observed variable and y_ϕ^{GS} is the corresponding class label in the retrieval database. Let $f(x_\phi^{GS}) : \{f_1(x_\phi^{GS}), f_2(x_\phi^{GS}), \dots, f_H(x_\phi^{GS})\}$ be the set of hidden layer features extracted for x_ϕ^{GS} GS-

ROBIN dataset sample, where $f_p(x_\phi^{GS})$ represents p^{th} hidden layer feature extracted from x_ϕ^{GS} sample in database and H represents the total number of hidden layers in deep feature hierarchy. For experimentation purpose, we have extracted features from 12 hidden layers of CNN model. For each sample in the database, features from each hidden layer ($f_p(x_\phi^{GS})$) are extracted and stored in the feature database as discussed in Sec. ???. In a similar manner, features are extracted from the query sketches as well. We compare the performance of individual hidden layer for proposed floor plan retrieval task. Let $f_s(q) : \{f_{s1}(q), f_{s2}(q), \dots, f_{sH}(q)\}$ be the set of hidden layer features extracted from a query sketch (q). Then, the Matching Score (M) between query sketch (q) and a sample from the database (x_ϕ^{GS}) using p^{th} hidden layer feature is calculated (and used for ranking), as:

$$M = \frac{|f_p(x_\phi^{GS})|}{\sum_{i=1}^{|f_p(x_\phi^{GS})|} |f_p^i(x_\phi^{GS}) - f_{s_p}^i(q)|_2} \quad (2)$$

Using Eq. 2, given a query sketch we obtain the similar generated sketches from the GS-ROBIN dataset. It is to be noted, that as these sketches have been generated from ROBIN dataset, therefore, transitively they correspond to the images they are generated from. Thus, through this indirect relation, we show the images belonging to [1] as the retrieved result given the query sketch.

III. EXPERIMENTAL SETUP

A. Implementation Details

We experimented with 2 datasets : (1) ROBIN dataset [1] and (2) S-ROBIN dataset. S-ROBIN dataset has 510 real world floor plan sketches divided into 3 broad categories and 17 sub-categories differing in the overall global outer shape of the layouts, designed in a similar fashion as ROBIN. For training the Cyclic GAN, we divided the datasets ROBIN and S-ROBIN into training (70%) and testing (30%) sets and performed repetition fold. Further, the Cyclic GAN with the help of the losses discussed in Sec. II-A, learns the mapping between both domains. We observed that visually the GAN generated sketches from ROBIN were of better quality. To make it clear to the reader, let query sketch be S , the ROBIN dataset images to be retrieved be $\{I\}$ and the generated sketches from ROBIN ($\{I\}$) be $\{GS\}$. Then as S and $\{GS\}$ are in the same domain, similar sketches in $\{GS\}$ are retrieved smoothly through REXplore. Further, it is to be noted these generated sketches $\{GS\}$ exactly correspond to their image counterparts from $\{I\}$. Hence, it is safe to say that through REXplore relationship between S and $\{GS\}$ has been determined. Also, relationship between $\{GS\}$ and $\{I\}$ pre-exists. Therefore, transitively, relationship between S and $\{I\}$ has thus been established. It is this indirect relationship that helps us justify that ROBIN dataset images similar to the query sketch can be shown as the retrieval result obtained through REXplore.



Figure 3. Top five rank ordered retrieval result of proposed framework for three different hand-drawn query floor plans.

For feature representation, we trained the CNN using the ROBIN dataset by dividing it into training set with 70% samples and test set with 30% samples. As a retrieval database, we have used the entire GS-ROBIN dataset which is generated from the ROBIN dataset and thus, transitively represents it. Each floor plan image has been resized to a size of 128×128 for both matching and feature representation task. For deep model configuration, we have followed the architecture described in [13].

IV. RETRIEVAL RESULTS

A. Qualitative Results

REXplore is able to successfully retrieve a rank order set of floor plan images from the ROBIN dataset given a query in the form of sketch, using the proposed domain mapping framework. The top 5 rank ordered retrieval results for three different hand-drawn queries, are shown in Fig. 3. The samples which are erroneously retrieved, i.e. not from the same class as that of the query class, are highlighted with a “red” bounding box. For all the results shown in Fig. 3, we have used the features extracted from the first normalization layer of the CNN stack, as it yields the best performance. The first rank ordered result retrieved from the ROBIN dataset, is the image exactly matching the query sketch. This is due to maximum matching score between the retrieved layout and the query layout. The subsequent results differ in the matching scores and are thus, ranked lower. In case of Fig. 3. (a), (b), all the top 5 retrieved samples belong to the query class. Global layouts of the all the floor plans are identical and so is the number of rooms in the floor plans. It is to be noted that there is a difference in the shape and position of the individual rooms, as well as, the furniture components for the rank ordered samples.

Justification of the Rank 5 erroneously retrieved result in Figure 3(c), can be understood by observing that the rooms in both the layouts share very similar adjacency. The bottom part of the floor plan is almost identical. Moreover, the size and type of furniture present in the individual rooms is quite similar in case of the query and the Rank 5 sample.

Hence, similarity score between the query floor plan and the retrieved Rank 5 layout is high.

B. Performance Evaluation

We quantify the performance of REXplore using Precision (P) and Recall (R) metric in Fig. 4. We have averaged the precision values over all the queries for the particular recall values. As discussed before, using CNN for the retrieval of floor plan images given a sketch is not straightforward (see the performance of [1] in Fig. 4). On the other hand, converting query sketches into corresponding image counterparts and then retrieving yielded weak performance. Therefore, sketch counterparts of ROBIN generated through Cyclic GAN are used for retrieval. To understand the effectiveness of the proposed CNN layers, we compared the PR and Mean Average Precision (MAP) of floor plan retrieval for individual layers. We analysed that normalized and pooling layers are quite powerful as compared to convolutional layer for floor plan retrieval task. We observed that the normalized layer 1 yielded the best average precision value of 0.63.

It can be observed that features extracted from the normalized layer 1 in case of REXplore (“blue” line) performs best with a MAP value of 0.63. Kindly note, the under-performance of the basic deep learning framework [1] (average precision value : 0.04393) using straight-forward feature extraction from CNNs. Performance of REXplore is also superior than the very recent technique [9] (improvement in average precision value by 0.24). Also, SIFT, HOG, Run Length Histogram and OASIS approach were observed to perform weakly as they are not able to capture the abstract and sparse nature of sketches and are unable to map between the features of images and sketches. Thus, the results justify the importance of the introduction of domain-mapping approach given sketch based query and image based retrieval samples.

V. CONCLUSION

In this paper, we have proposed a deep learning framework using GAN model for sketch based retrieval of building

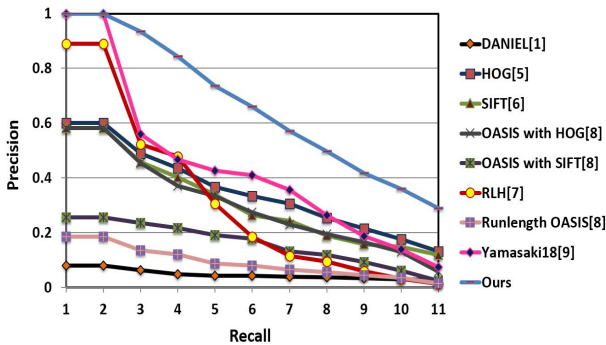


Figure 4. PR plot, comparing state-of-art methods with REXplore.

floor plan images. We are able to achieve an average precision value of 0.63 upon experimenting real world floor plan images. In future, we are planning to explore the possibilities of incorporating better domain adaptation techniques and look into the partial/abstract floor plan specifications to bridge the gap between sketch and scanned floor plans and achieve superior retrieval performance.

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